

FP2 Chapter 12 Differential Equations

Lecture 12 Differential Equations

(First Order Linear DE, Second Order Linear DE, Non-homogenous linear DE, Substitution)

Introduction:

- In P3, you have encounter differential equation chapter. You were mainly solving first order linear differential equations with separable variables.
- In this chapter, we will now consider three other main types of differential equation:
 - (i) First order linear differential equation requiring an **integrating factor**
 - (ii) **Second order linear differential equations** with constant coefficients
 - (iii) Solution of differential equations by **substitution**
- Each of these terms has a meaning:
 - (i) First order: Contain the term $\frac{dy}{dx}$.
 - (ii) Second order: Contain the term $\frac{d^2y}{dx^2}$.
 - (iii) Linear: There are no product of derivatives in the equations.
 - (iv) Constant coefficients: Each derivative is multiplied by, at most, a constant.

1. First Order Linear Differential Equations requiring an Integrating Factor

- The first order linear differential equation is of the form $\frac{dy}{dx} + Py = Q$ where P and Q are functions of x .
- Such an equation can be solved by first multiplying both sides by the **integrating factor** $e^{\int P dx}$.

$$\frac{dy}{dx} + Py = Q$$

$$e^{\int P dx} \frac{dy}{dx} + P e^{\int P dx} y = Q e^{\int P dx}$$

- Since the left-hand side is the differential of $y e^{\int P dx}$, we therefore have

$$\frac{d}{dx} (y e^{\int P dx}) = Q e^{\int P dx}$$

which gives

$$y e^{\int P dx} = \int Q e^{\int P dx} dx$$

- The right hand side is often integrated by parts.

- **Example 1:**

Show that the differential equation $\frac{dy}{dx} + Py = Q$, where P and Q are functions of x , can be solved by use of an integrating factor, and give the integrating factor.

Hence solve the differential equation $x \frac{dy}{dx} + y = e^x$.

- **Example 2:**

Solve the differential equation $\frac{dy}{dx} + 2y \tan x = \sin x$ given that $y = 1$ when $x = \frac{\pi}{2}$.



- **Example 3:**

Solve the differential equation $\frac{dy}{dx} + 2y = x^2$.

Sunway A Levels

- **Example 4:**

Solve the differential equation $(1+x)\frac{dy}{dx} - y + 2x + x^2 = 0$, given that $y = 0$ when $x = 0$.

Sunway A Levels

- **Example 5:**

Solve the differential equation $\frac{dy}{dx} - 3y \tanh x = 1$, given that $y = 0$ when $x = 0$.

(Hint: Use $\cosh 3x = 4 \cosh^3 x - 3 \cosh x$)

Sunway A Levels

- **Example 6:**

Find y in terms of x , given that $x \frac{dy}{dx} + (1 + x \tanh x)y = 1$ and that $y = 0$ when $x = \ln 2$.

Sunway A Levels

2. Second Order Linear Differential Equations with Constant Coefficients

Linear Differential Equations

- Before solving second order differential equations you will find it helpful to look at the form of the solution of **first order linear homogenous differential equations with constant coefficients**. The general first order equation of this type can be written as $a \frac{dy}{dx} + by = 0$.
- **First order because the highest derivative is 1, homogenous because the RHS is 0, constant coefficients because a and b are constants.**
- You can solve this general differential equation using the method of separation of variables:

$$\frac{dy}{dx} = -\frac{b}{a}y$$

$$\int \frac{1}{y} dy = \int -\frac{b}{a} dx$$

$$\ln |y| = -\frac{b}{a}x + c$$

- This gives $y = \pm e^{(-ax+c)} \Rightarrow y = Ae^{-\frac{b}{a}x}$, where $A = \pm e^c$ is a constant.
- **This shows that the solution of any first order equation of this type will contain an exponential function and an unknown constant.**
- Knowing the form of the solution allows you to find it **without doing any integration**.
- This is demonstrated in the next example. Solve the differential equation $a \frac{dy}{dx} + by = 0$.
- We begin by assuming an exponential solution of the form $y = Ae^{mx}$, where A and m are constants. If this is to be a solution, it must satisfy the differential equation.

- Differentiating y with respect to x , $\frac{dy}{dx} = mAe^{mx}$. Substituting this into the differential equation gives $mAe^{mx} + bAe^{mx} = 0 \Rightarrow am + b = 0 \Rightarrow m = -\frac{b}{a}$. So the general solution of the differential equation is $y = Ae^{-\frac{b}{a}x}$.
- The equation $am + b = 0$ is called the **auxiliary equation** and this method is called the **auxiliary equation method**. We shall extend this concept to second order differential equations.
- A **second order linear differential equation with constant coefficients** can be written in the form $a\frac{d^2y}{dx^2} + b\frac{dy}{dx} + cy = f(x)$, where a, b and c are constants.
- The equation is **homogenous** if $f(x) = 0$. Otherwise it is non-homogenous.
- In this section, we will investigate the homogenous case of this equation:

$$a\frac{d^2y}{dx^2} + b\frac{dy}{dx} + cy = 0$$

- The following is a complete proof for the solution to this differential equation
- Suppose that we assume a solution of the form $y = Ae^{mx}$, then

$$\frac{dy}{dx} = Ame^{mx}, \frac{d^2y}{dx^2} = Am^2e^{mx}$$
- Substituting these into the differential equation gives $am^2 + bm + c = 0$, this is the auxiliary equation or simply the **auxiliary quadratic equation (ACE)**.
- **Notice that the form of the AQE is very close to that of the original differential equation. It could have been written down straight away without any of the intermediate working.**
- Since AQE has two roots, real or complex, there are two solutions of the differential equation. For example, suppose $m = m_1, m_2$, then $y = Ae^{m_1x}$ and $y = Be^{m_2x}$ (where A and B are constants) are two solutions of the differential equation.
- Therefore, the general solution of the differential equation $a\frac{d^2y}{dx^2} + b\frac{dy}{dx} + cy = 0$ is $y = Ae^{m_1x} + Be^{m_2x}$.

- The expression $Ae^{m_1x} + Be^{m_2x}$ is usually called simply the **complementary function** (y_c).

- **Case 1: AQE with two real distinct roots**

If $m_1 = m_2$, then the complementary function is $y = Ae^{m_1x} + Be^{m_2x}$.

- **Case 2: AQE with repeated roots**

If $m = m_1$, then you have only one solution $y = Ae^{m_1x}$. But since the equation was of second order, you expect the AQE to have two roots, giving the general solution with two unknown constants.

However, it can be shown that $y = Bxe^{m_1x}$ is also a solution of the differential equation.

Therefore, the complementary function is $y = Ae^{m_1x} + Bxe^{m_1x}$.

- **Case 3: AQE with complex roots.**

If $m = p \pm iq$, then the complementary function is $y = Ce^{(p+iq)x} + De^{(p-iq)x}$

$$y = e^{px}(Ce^{iqx} + De^{-iqx})$$

$$y = e^{px}(C \cos qx + C i \sin qx + D \cos qx - D i \sin qx)$$

$$y = e^{px}[(C + D) \cos qx + i(C - D) \sin qx]$$

$$y = e^{px}(A \cos qx + B \sin qx)$$

- **Example 7:**

Find the general solution of each of these second order homogenous differential equations.

(i) $\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = 0$

(ii) $\frac{d^2y}{dx^2} - 4y = 0$

(iii) $\frac{d^2x}{dt^2} - x = 0$

(iv) $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 2y = 0$

(v) $\frac{d^2y}{dx^2} + 10\frac{dy}{dx} + 25y = 0$

(vi) $\frac{d^2y}{dx^2} - 9y = 0$

(vii) $\frac{d^2y}{dx^2} - 8\frac{dy}{dx} + 16y = 0$

(viii) $\frac{d^2x}{dt^2} - 4\frac{dx}{dt} + 5x = 0$

(ix) $\frac{d^2y}{dx^2} - 8\frac{dy}{dx} + 15y = 0$

Non-Homogenous Linear Differential Equations

- Consider the general second order linear non-homogenous differential equation

$$a \frac{d^2 y}{dx^2} + b \frac{dy}{dx} + cy = f(x).$$

- As a first step, look at the first order non-homogenous linear differential equation

$$\frac{dy}{dx} + 2y = 3x - 1$$

$$e^{2x} \frac{dy}{dx} + 2e^{2x} y = e^{2x} (3x - 1)$$

$$\frac{d}{dx} (e^{2x} y) = e^{2x} (3x - 1)$$

$$e^{2x} y = \int e^{2x} (3x - 1) dx$$

$$e^{2x} y = \frac{3}{2} x e^{2x} - \frac{5}{4} e^{2x} + A$$

$$y = A e^{-2x} + \left(\frac{3}{2} x - \frac{5}{4} \right)$$

- Notice that the solution consists of two parts.
 - (i) The first part $A e^{-2x}$, contains an arbitrary constant, A , and satisfies the homogenous equation, $\frac{dy}{dx} + 2y = 0$. It is the complementary function of the equation formed by replacing right hand side with zero.
 - (ii) The second part $\left(\frac{3}{2} x - \frac{5}{4} \right)$, includes no arbitrary constants.
- This suggests that part of the solution depends only on the LHS of the differential equation, and part on the full equation. To explore this idea further, look carefully the next example which has the same LHS as the previous one, but a different RHS.

$$\frac{dy}{dx} + 2y = e^{3x}$$



$$e^{2x} \frac{dy}{dx} + 2e^{2x}y = e^{5x}$$

$$\frac{d}{dx}(e^{2x}y) = e^{5x}$$

$$e^{2x}y = \int e^{5x} dx$$

$$e^{2x}y = \frac{1}{5}e^{5x} + A$$

$$y = Ae^{-2x} + \frac{1}{5}e^{3x}$$

- This is the same complementary function Ae^{-2x} and another part, in this case $\frac{1}{5}e^{3x}$, which contains no arbitrary constants.
- The complementary function is the same in each of the above two examples, and could have been found by solving the corresponding homogenous equation. The other part is the same sort of function as the RHS of the differential equation. This function is called the **particular integral** (y_p).
- How to find a particular integral? The first step is to write down a suitable **trial function**. The form of this trial function will be determined by the RHS of the differential equations. This is called **method of undetermined coefficients**.
- You may have four types of RHS: polynomial function, trigonometry function, exponential function and hyperbolic function. You could have any sum or product of these. For examples,

RHS	Trial Function
$x^2 + 2x$	$Ax^2 + Bx + C$
$\sin 3x$	$A \sin 3x + B \cos 3x$
e^{4x}	Ae^{4x}
$8 \sinh x$	$A \sinh x + B \cosh x$



$x^2 + 2x + e^{4x}$	$Ax^2 + Bx + C + De^{4x}$
$xe^{2x} \cos 4x$	$(Ax + B)e^{2x} \cos 4x + (Cx + D)e^{2x} \cos 4x$

- From trial function, find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ substitute into the question to find the value of A, B, C etc. This will be the particular integral (y_p).
- **Principle of Superposition:** If y_c is a solution to $a \frac{d^2y}{dx^2} + b \frac{dy}{dx} + c = 0$, and y_p is a solution to $a \frac{d^2y}{dx^2} + b \frac{dy}{dx} + c = f(x)$, then $y_c + y_p$ is also a solution to $a \frac{d^2y}{dx^2} + b \frac{dy}{dx} + c = f(x)$.
- The **particular solution** of linear non-homogenous differential equation can be obtained if the initial conditions are given.

- **Example 8:**

Find the general solutions / particular solutions of each of these second order non-homogenous differential equations. Your results from **Example 7** will be useful here.

(i) $\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = 2x + 5$

(ii) $\frac{d^2y}{dx^2} - 4y = x$



(iii) $\frac{d^2x}{dt^2} - x = e^{-2t}$

(iv) $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 2y = \sin 3x$

(v) $\frac{d^2y}{dx^2} + 10\frac{dy}{dx} + 25y = 441 \cosh 2x$

(vi) $\frac{d^2y}{dx^2} - 9y = 18x^2 - 5 + 8 \sinh x$

(vii) $\frac{d^2y}{dx^2} - 8\frac{dy}{dx} + 16y = 32x^2 - 28$, given that $y = 2, \frac{dy}{dx} = 9$ when $x = 0$

(viii) $\frac{d^2x}{dt^2} - 4\frac{dx}{dt} + 5x = 20(t + e^{-t})$, given that $x = 5, \frac{dx}{dt} = 2$ when $t = 0$.

(ix) $\frac{d^2y}{dx^2} - 8\frac{dy}{dx} + 15y = 130 \sin x$, given that $y = 12, \frac{dy}{dx} = 23$ when $x = 0$.

Sunway A Levels

More Problems (Special Case):

- If the complementary function contains elements that you would like to include in the trial solution, multiply the trial solution by x^m such that m is the smallest integer so that the resulting term is different from any of the complementary function.
- **Example 9:**

Find the general solution to the following differential equations

(i) $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = 4e^{2x}$



(ii) $\frac{d^2y}{dx^2} + 9y = 18 \sin 3x$

Sunway A Levels



(iii) $\frac{d^2y}{dx^2} + 3\frac{dy}{dx} - 4y = 5e^x$

Sunway A Levels



(iv) $\frac{d^2y}{dx^2} - 8\frac{dy}{dx} + 15y = 4e^{5x}$

Sunway A Levels



(v) $\frac{d^2y}{dx^2} - 9\frac{dy}{dx} = 18e^{9x}$

Sunway A Levels



(vi) $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = 4e^{2x}$

Sunway A Levels

Problem Solving:**- Example 10 (PYQ O/N 2012/13):**

Obtain the general solution of the differential equation

$$\frac{d^2x}{dt^2} + 6\frac{dx}{dt} + 13x = 75 \cos 2t. \quad [7]$$

Given that $x = 5$ and $\frac{dx}{dt} = 0$ when $t = 0$, find x in terms of t . [4]

Show that, for large positive values of t and for any initial conditions,

$$x \approx 5 \cos(2t - \phi),$$

where the constant ϕ is such that $\tan \phi = \frac{4}{3}$. [3]

- **Example 11 (PYQ O/N 2011/13):**

- 6 Find the general solution of the differential equation

$$\frac{d^2x}{dt^2} + 4\frac{dx}{dt} + 4x = \sin 2t. \quad [6]$$

Describe the behaviour of x as $t \rightarrow \infty$, justifying your answer. [2]

- **Example 12 (PYQ O/N 2002):**

- 8 The value of the assets of a large commercial organisation at time t , measured in years, is $\$(10^8 y + 10^9)$. The variables y and t are related by the differential equation

$$\frac{d^2 y}{dt^2} + 5 \frac{dy}{dt} + 6y = 15 \cos 3t - 3 \sin 3t.$$

Find y in terms of t , given that $y = 3$ and $\frac{dy}{dt} = -2$ when $t = 0$. [9]

Show that, for large values of t , the value of the assets is less than $\$9.5 \times 10^8$ for about a third of the time. [3]

- **Example 13 (PYQ O/N 2003):**

- 7 Find the general solution of the differential equation

$$\frac{d^2y}{dt^2} + 4\frac{dy}{dt} + 4y = e^{-\alpha t},$$

where α is a constant and $\alpha \neq 2$.

[7]

Show that if $\alpha < 2$ then, whatever the initial conditions, $ye^{\alpha t} \rightarrow \frac{1}{(2-\alpha)^2}$ as $t \rightarrow \infty$.

[2]

3. Solution of Differential Equations by Substitution

- There are some differential equations, which, while not of the correct form, can be transformed into the correct form by substitution.

- **Example 14:**

By using the substitution $v = y - x$, or otherwise, find the general solution of the differential equation $(y - x) \frac{dy}{dx} = y$, expressing x as a function of y .

- **Example 15:**

The variables x and y are related by the differential equation $\frac{dy}{dx} = \frac{2x+y+2}{2x+y-1}$. Using the substitution $z = 2x + y$, show that $\frac{dz}{dx} = \frac{3z}{z-1}$. Solve the equation and hence find the general solution of the given differential equation.

Sunway A Levels

- **Example 16:**

Using the substitution $u = y^2$, or otherwise, find the general solution of the differential equation $2xy \frac{dy}{dx} = y^2 - 4x^2$.

Sunway A Levels

- **Example 17:**

By means of the substitution $z = \frac{1}{y^2}$, transform the differential equation $\frac{dy}{dx} + y = xy^3$ into a differential equation containing z and x . Solve this equation for z and hence show that the solution of the given differential equation for which $y = \sqrt{2}$ when $x = 0$ is $y^2 = \frac{2}{2x+1}$.

Sunway A Levels

- **Example 18 (PYQ M/J 2015/11/12):**

Show that the substitution $v = \frac{1}{y}$ reduces the differential equation

$$\frac{2}{y^3} \left(\frac{dy}{dx} \right)^2 - \frac{1}{y^2} \frac{d^2y}{dx^2} - \frac{2}{y^2} \frac{dy}{dx} + \frac{5}{y} = 17 + 6x - 5x^2$$

to the differential equation

$$\frac{d^2v}{dx^2} + 2 \frac{dv}{dx} + 5v = 17 + 6x - 5x^2. \quad [4]$$

Hence find y in terms of x , given that when $x = 0$, $y = \frac{1}{2}$ and $\frac{dy}{dx} = -1$. [10]

- **Example 19 (PYQ O/N 2009/1):**

- 9 Show that if y depends on x and $x = e^u$ then

$$x^2 \frac{d^2 y}{dx^2} = \frac{d^2 y}{du^2} - \frac{dy}{du}. \quad [4]$$

Given that y satisfies the differential equation

$$x^2 \frac{d^2 y}{dx^2} + 5x \frac{dy}{dx} + 3y = 30x^2,$$

use the substitution $x = e^u$ to show that

$$\frac{d^2 y}{du^2} + 4 \frac{dy}{du} + 3y = 30e^{2u}. \quad [2]$$

Hence find the general solution for y in terms of x . [5]

- **Example 20 (PYQ M/J 2003/1):**

- 9 The variables x and t , where $x > 0$ and $0 \leq t \leq \frac{1}{2}\pi$, are related by

$$x \frac{d^2x}{dt^2} + \left(\frac{dx}{dt} \right)^2 + 5x \frac{dx}{dt} + 3x^2 = 3 \sin 2t + 15 \cos 2t,$$

and the variables x and y are related by $y = x^2$. Show that

$$\frac{d^2y}{dt^2} + 5 \frac{dy}{dt} + 6y = 6 \sin 2t + 30 \cos 2t. \quad [3]$$

Hence find x in terms of t , given that $x = 2$ and $\frac{dx}{dt} = -\frac{3}{2}$ when $t = 0$. [10]

- **Example 21 (PYQ O/N 2007/1):**

Show that the substitution $y = \frac{1}{w}$ reduces the differential equation

$$y \frac{d^2 y}{dx^2} + 2y \frac{dy}{dx} - 2 \left(\frac{dy}{dx} \right)^2 - 5y^2 = (5x^2 + 4x + 2)y^3$$

to

$$\frac{d^2 w}{dx^2} + 2 \frac{dw}{dx} + 5w = -5x^2 - 4x - 2. \quad [4]$$

Find the general solution for w in terms of x . [6]

Find a function f such that $\lim_{x \rightarrow \infty} \left(\frac{y}{f(x)} \right) = 1$. [3]